

NeurIPS 2023: A Sharp, Two-Sided Test-Error Bound for Finite-Rank KRR

Freeze a pretrained network, retrain only its last layer, and you are doing finite-rank kernel ridge regression — a regime where the textbook generalization bounds go loose. This paper makes them tight in both directions, for any amount of regularization.

Here is a recipe you have almost certainly used: take a pretrained network, freeze the backbone, and retrain only the final linear layer. It is fast, cheap, and everywhere — from few-shot transfer to reservoir computing. What is less obvious is that this recipe is exactly kernel ridge regression (KRR) with a finite-rank kernel. The frozen features define a fixed, finite-dimensional feature map, and fitting the last layer is ridge regression in that feature space, whose kernel has rank equal to the number of frozen features it exposes.

A natural question follows: how large is the test error of that last layer, and can we certify it before ever touching the test set? Statistical learning theory has answers for KRR in general, but in the finite-rank regime those answers are loose. Classical bounds are frequently one-sided (upper bounds only), several require the ridge parameter λ to stay bounded away from zero, and some become vacuous precisely in the ridgeless limit $\lambda \rightarrow 0$ that last-layer training tends to approach. Without a matching lower bound, you also cannot tell whether a small upper bound is genuinely small or merely optimistic.

This NeurIPS 2023 paper closes the gap. For finite-rank KRR in the under-parameterized regime — sample size N larger than the kernel rank M — the authors derive matching non-asymptotic, high-probability upper and lower bounds on the test error. The bounds stay tight for any ridge λ , including the ridgeless limit, and empirically the new upper bound tracks the true error where the closest prior bound floats orders of magnitude above it.

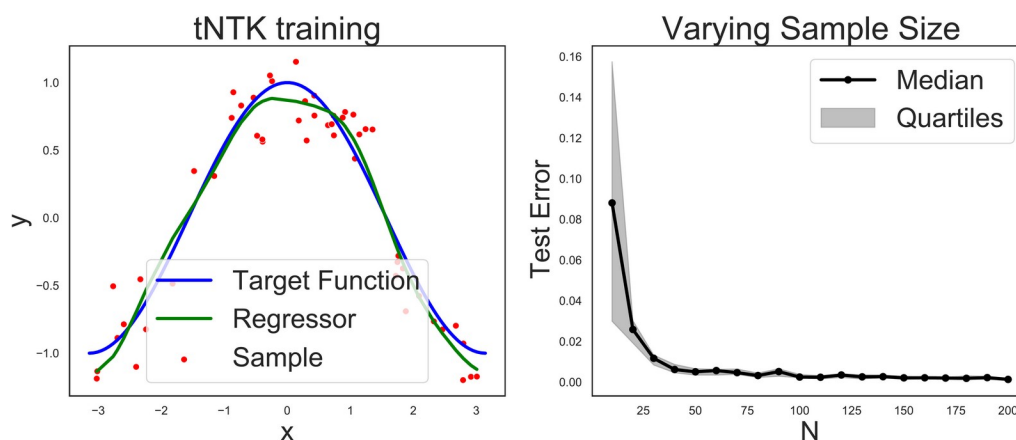


Figure 1. Finite-rank KRR on a truncated neural tangent kernel (tNTK). Left: with $N=50$ samples and rank $M=7$, the regressor (green) tracks the target (blue) despite noisy samples (red). Right: median test error over 10 trials drops steeply as N grows from 10 to 200 (shaded interquartile bands).

Paper: <https://arxiv.org/abs/2310.00987> (NeurIPS 2023). Authors: Tin Sum Cheng, Aurelien Lucchi, Ivan Dokmanić, Anastasis Kratsios, and David Belius.

Why the finite-rank regime breaks the textbook bounds

A finite-rank kernel spans a finite-dimensional space inside the reproducing-kernel Hilbert space (RKHS), with an eigenbasis of M functions. The target the model must learn rarely lives entirely in that span. It splits into an in-span part, which finite-rank KRR can in principle recover, and an orthogonal complement that the estimator can never represent. That complement is an irreducible finite-rank error: a floor on the test error that no amount of extra data removes, and a term the analysis has to track honestly rather than wish away.

Prior finite-rank analyses each miss part of this picture. Some hold only in asymptotic, high-dimensional limits; others are valid only when λ decays on a prescribed schedule tied to the sample size N ; others bound only the expected error rather than a single realized run; and almost none supply a lower bound. A one-sided, asymptotic, schedule-dependent guarantee is not what you want when you have one frozen backbone, one dataset, and one value of λ in front of you.

A bias-variance split, analyzed in the eigenfunction basis

The starting point is the ordinary KRR estimator — the minimizer of a regularized least-squares objective over the RKHS:

$$\hat{f}_{\{Z, \lambda\}} = \operatorname{argmin}_{f \in H} (1/N) \sum_i (f(x_i) - y_i)^2 + \lambda \|f\|_H^2.$$

The test error is the expected squared deviation from the true target, and it splits cleanly into a bias term (the error with noiseless labels) and a variance term (driven by label noise). The authors bound each piece separately, with high probability. The move that makes the constants sharp is a change of coordinates: instead of working in the RKHS basis, they analyze the target in the L^2 eigenfunction basis $\{\psi_k\}$. This decouples the kernel spectrum from the randomness of where the samples happen to land, which is precisely the coupling that inflates earlier bounds.

Two technical ingredients carry the proof: a sub-Gaussian concentration result for the random covariance matrix, and a Neumann-series expansion of the matrix inverse that appears in the ridge terms. Together they let the bound hold for any λ chosen independently of N — not just for a ridge that must shrink as the sample grows. Concretely, the variance obeys a bound of the form

$$\text{variance} \leq \sigma^2 \cdot (2M/N) \cdot (1 + \sqrt{(\log N / N)}) + C_2 \cdot (\log N / N),$$

so it scales like the noise variance times twice the rank over the sample size, with residue terms that fade at a $\log N / N$ rate. That decay is an order of magnitude faster than the $O(\sqrt{(\log N / N)})$ rate a generic Rademacher-complexity argument would deliver here.

The payoff: bounds that actually bracket the error

The most visible gain is against the closest prior result, due to Bach. The figure below plots both upper bounds on a natural-log scale, once as the sample size N varies and once as the

inverse ridge $1/\lambda$ varies. The new bound sits far below Bach's across the board, and — crucially — it keeps improving toward the ridgeless direction (large $1/\lambda$) where Bach's bound turns around and gets worse. That contrast isolates the ridge dependence as the main source of the improvement.

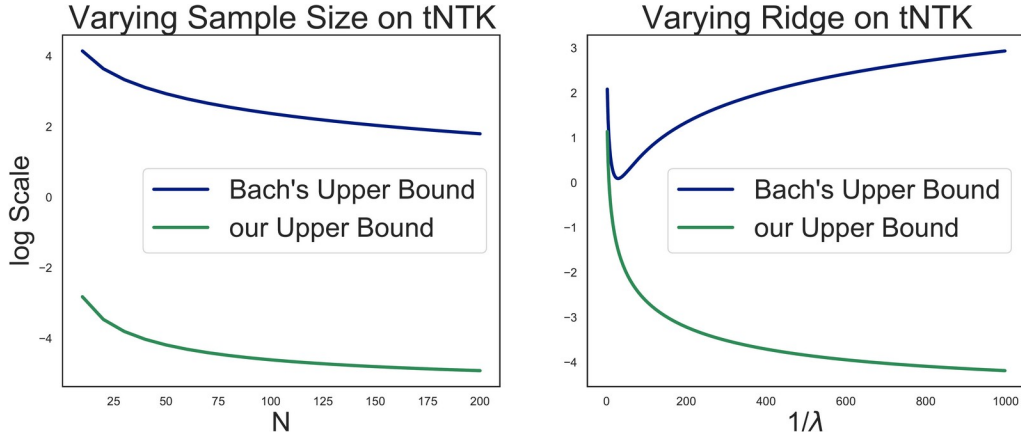


Figure 2. New upper bound (green) vs Bach's upper bound (blue) on the tNTK, natural-log scale. Left: varying N . Right: varying inverse ridge $1/\lambda$. The new bound is orders of magnitude lower and keeps improving as regularization vanishes, where Bach's turns worse.

A lower bound is what turns a good upper bound into a tight one. The next figure overlays the paper's upper and lower bounds (with the small residue terms dropped for clarity) on the measured test error for two controlled finite-rank kernels — the tNTK and a separately constructed low-rank kernel (LK). The two bounds sandwich the median error across the range of N and squeeze together as N grows, which is the empirical signature of tightness. Because the residues are omitted, the simplified bounds stop bracketing the error at very small N , exactly where the theory says those terms still matter; they recover their grip as the sample grows and the small residues steadily fade away.

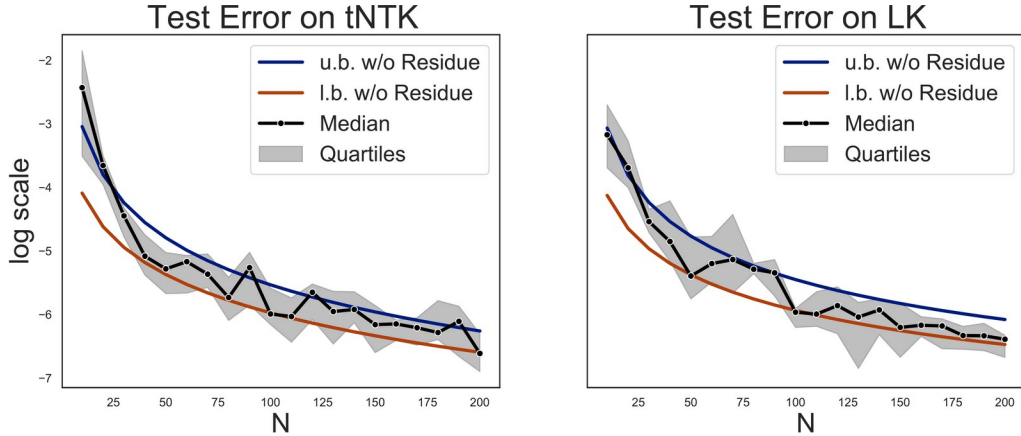


Figure 3. Upper and lower bounds (residues dropped) vs measured test error over 10 trials, on the tNTK (left) and low-rank kernel LK (right), natural-log scale. The bounds bracket the median error and converge as N grows; the small- N gap is expected.

How it compares, at a glance

Table 1. What the new analysis adds over prior finite-rank KRR bounds.

Property	Prior finite-rank bounds	This paper
Upper bound	Yes	Yes
Lower bound	Rarely	Yes, matches the upper bound as N grows
Ridgeless limit $\lambda \rightarrow 0$	Often vacuous	Non-vacuous
Non-asymptotic, fixed λ	Not always	Yes, any λ independent of N
Upper-bound decay rate	up to $\sqrt{(\log N / N)}$	$\log N / N$

What it means, and where it stops

The practical reading is that finite-rank KRR — the model behind last-layer fine-tuning — now comes with an honest, two-sided certificate of its test error. The bounds hold with high probability (at least $1 - 2/N$ over the random sampling), they stay tight for any ridge including none at all, and they make the irreducible finite-rank floor explicit rather than hiding it. The enabling idea — analyze the target in the eigenfunction basis to separate spectrum from sampling noise — is a template other finite-rank analyses can borrow.

The honest boundary: the results are for the under-parameterized regime where N exceeds the rank M , and the validation uses controlled synthetic kernels (tNTK and LK) where the spectrum and target decomposition are known, not natural-image datasets. In the ridgeless limit the analysis further shows the bias is governed purely by the finite-rank error and the variance purely by the noise level σ^2 . Within those limits, this is a much sharper certificate than the loose, one-sided, asymptotic guarantees that came before.